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**Scheduling Demand Response
Resources Under Net-load Uncertainty
Using Multi-stage Stochastic
Programming**

by

Michael Tiller

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Supervised by Professor Miguel Anjos

Abstract

An increasing volatile net-load profile, driven by growing renewable penetration and the growth of large uncertainty flexible loads such as AI data centres, is placing pressure on utilities to incorporate sources of flexibility into their scheduling decisions. Demand response (DR) provides one source of demand side flexibility, however DR is mostly incorporated in scheduling problems as a continuous, price-elastic variable. This assumption allows these models excess freedom to curtail or shift demand than real contracted DR products typically allow, since most DR programmes activate pre-agreed, fixed-shape load reductions rather than allow for real-time continuous load adjustments.

This dissertation develops a multi-stage stochastic programme that instead represents DR resources as a fixed set of enumerated start-time and duration profiles, which include both the interruptible period and the subsequent rebound in consumption. DR resources are then activated through a binary decision variable. Day-ahead (DA) resources are scheduled within the first stage, while real-time (RT) resources are scheduled in the second stage, conditioned on the realisation of uncertainty at a midday decision point. The model is evaluated across four uncertainty cases representing different sources of net-load risk, a base case (day ahead prediction uncertainty), an AI-driven demand spike, a combined spike and peak-shift case, and a net load case incorporating wind generation uncertainty.

Two questions are answered using this approach. The first is whether adopting a stochastic approach is worth its added computational cost, tested using the Value of the Stochastic Solution and the Expected Value of Perfect Information. A VSS of 5.8 to 7.7% in the higher uncertainty cases is enough to justify a stochastic approach in every case except the base case. The second is the value of flexibility within a DR resource, specifically the granularity of allowed start times and the length of the interruptible period. Restricting either has a significant effect on scheduling cost, the size of the peak, and ramping. The model also reveals a clear distinction in the roles of the two resource types. Day-ahead resources are used mainly to reduce the size of the peak, while real-time resources are mainly used to target ramping, which explains why start-time granularity matters far more for real-time resources, coarser start times erode their ability to target ramps precisely. This effect is especially pronounced in the wind-driven case, where hourly-shaped DR profiles are a poor match for wind generation that does not move in neat hourly steps.

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Own Work Declaration

I declare all work is my own, except for any instances where I have indicated so within the text.

AI Declaration

Project studies: Generative AI was additionally used in the following ways in this project.

- Claude, July-August 2026, was used to help with graph formatting within the matplotlib Python library.
- Claude, August 2026, was used to help brainstorm the incorporation of wind generation into the scenario tree, and to help debug errors with implementing this into the scenario generation process code.
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1 Introduction

The increasing penetration of Renewable Energy within the grid has created a need to additional flexibility within the grid network to cope with the variability in the generation of electricity. There are two distinct solutions to absorb this variable generation: productions-side flexibility and demand-side flexibility. One form of demand side flexibility is that of Demand Response (DR), DR helps utilities manage peak demand and maintain grid reliability, while also reducing customers' energy use and utility bills. DR programs are widely used across the industry, typically taking one of two forms: incentive-based programs, which reward customers for reducing consumption on request, and price-based programs, such as time-of-use tariffs. One specific incentive-based structure is the Direct Load Control Contract (DLCC), under which a utility calls on enrolled customer groups to reduce their consumption for a fixed amount and a fixed duration. As an example, participating customers under Southern California Edison's DLCC program can be called at any time across the year, subject to limits of once per day, ten times per calendar month, and a total interruption time of 180 hours per year [15].

This dissertation schedules DR resources that can be thought of as similar to DLCCs, however assumes these resources are supplied to the utility by a DR aggregator. A DR aggregator is an intermediary that pools DR resources from multiple sources and offers it to the utility as a larger resources with greater capacity, rather than contracted directly with individual customers as with DLCC's. This approach allows us to follow a discrete, fixed-profile representation of DR resources rather than the continuous, price-elastic recourse representation common elsewhere in the literature [7, 9]. The approach followed within this dissertation allows the utility to make a day-ahead decision, choosing which fixed-profile DR resources to call based on its forecast of the next days load profile. It can then make further decisions at a partially resolved noon forecast, over which flexible resources to call. Four conditions are imposed on how these resources may be called:

- Fixed day-ahead resources must be committed the day before any realisation of the load profile.
- Flexible resources can be committed on the day of the interruption, however since the interruption occurs in the afternoon, the decision must be made by noon.
- The length of the interruptible period cannot exceed a predetermined value.
- No DR resource can be called more than once per day.

The forecast load profiles are represented as vectors in $\mathbb{R}_+^T$, where T is the number of periods in the day, this dissertation uses a period length of 5 minutes, so $T = 288$. These load profiles vary significantly with the time of year, where demand is typically bimodal in winter, driven by heating-related peaks in the morning and evening, and unimodal in summer. Since demand response is only used during periods of peak stress, a single representative day with a pronounced peak is used as the baseline load profile for this study, from which additional uncertainty is added to the baseline profile.

The dissertation has two principal objectives: (i) quantifying the value of incorporating uncertainty explicitly into the scheduling problem, and (ii) quantifying the operational value of increased flexibility within demand response contract structures.

The Benefit of Incorporating Stochasticity

The literature on demand response scheduling contains both deterministic and stochastic optimisation models, however the value of adopting the more complex approach is unclear. Especially since day-ahead demand forecasts are often highly accurate, it is not immediately evident whether the additional computational complexity of a stochastic formulation is justified. This dissertation therefore evaluates the benefit of stochastic optimisation using the Value of the Stochastic Solution (VSS) and the Expected Value of Perfect Information (EVPI). VSS measures the cost of committing day-ahead demand response resources based on an expected-value forecast rather than a full scenario representation, while EVPI quantifies the maximum benefit obtainable if future uncertainty were known perfectly in advance, therefore isolating the impact of uncertainty. Furthermore, with increasing load from AI

data centres and the resulting uncertainty in demand, accurately representing this uncertainty and understanding its impacts is becoming increasingly important for power system operation. This is discussed further in section 2.7 and investigated in section 4 where the value of stochastic scheduling is expected to increase under greater variability in the timing of demand peaks. Another, source of uncertainty comes from the increasing penetration of renewable generation, in this case wind, which introduces additional variability into the net load profile that is not present in demand alone. This dissertation examines this by considering a load profile that is net of wind generation, as a further uncertainty scenario which is described in Section 3.3.3. Together, these sources of uncertainty allow the benefit of stochastic scheduling to be assessed not just as a function of how much uncertainty is present, but of where it originates, whether from demand-side timing shifts or from supply-side generation variability. This analysis establishes the conditions under which the additional complexity of a stochastic formulation is justified relative to a deterministic approach.

The Benefit of Flexibility Within DR Contracts

The second objective concerns the representation of DR resources. While the majority of the existing literature models demand response as a continuous, price-responsive recourse variable [7, 9], this dissertation instead represents demand response events as discrete, fixed-shape response profiles that can only be activated at a finite set of enumerated start times. This formulation more closely reflects the operational constraints of utilities, where they are given a fixed interpretable profile including the expected rebound, subject to specific operational constraints such as callable times and number of callable contracts (Hydro-Québec, personal communication, 2026).

The dissertation also looks at how much scheduling flexibility actually matters in practice, comparing hourly against 5-minute start-time windows to see where extra timing precision stops improving peak demand and ramping. Alongside this, the length of the interruptible period is varied too, to see how much longer DR events are actually worth and which contract features matter most for system performance.

The rest of the dissertation is organised as follows. Section 2 reviews the existing literature on demand response modelling, uncertainty representation, and DLCC scheduling, and uses this to position the dissertation’s two contributions. The methodology, covering the modelling choices, scenario generation, and the multi-stage stochastic formulation, is then set out in Section 3. Results follow in Section 4, covering both the benefit of stochasticity across the four uncertainty cases and the benefit of scheduling flexibility for DA and RT resources. Finally, Section 6 draws the work together, discussing its limitations and the conclusion.

2 Literature Review

A significant proportion of the literature incorporates demand response as part of their model as a method to provide additional flexibility within commitment or grid operational problems. The way in which DR is modelled varies significantly between papers, with some modelling DR as a continuously adjustable price-elastic resource, while others model DR as discrete resources which are subject to predefined limits on when and how they can be activated. This distinction is important as the amount of flexibility given to the DR resource within the model may not necessarily represent the flexibility available through a real-world DR contract. In addition, many of these operational problems contain uncertainty in demand, renewable generation or electricity prices, leading to a further distinction between deterministic and stochastic approaches to DR scheduling.

2.1 Continuous/Price Elastic DR Formulations

[7] incorporates DR into a stochastic two stage day-ahead optimal bidding problem for a micro-grid, where the objective is to minimise the expected operating costs. DR is modelled through flexible load that can be either curtailed or scheduled based on an associated marginal benefit within the second stage for scheduling the load. The benefit of DR is modelled using a quadratic benefit function derived from the price elasticity of demand, while the formulation of price-responsive shiftable

demand is adopted from [8]. In addition they also model flexible demand as spinning reserve bids. To address operational uncertainty, the authors propose both a two-stage stochastic optimisation model, which captures uncertainty in electricity prices and intermittent renewable generation, and a robust optimisation model that enables the level of conservatism towards real-time price uncertainty to be controlled. However, load uncertainty is not modelled explicitly. Instead, its effects are assumed to be implicitly captured within the other sources of uncertainty. A limitation of this DR formulation is that total demand is divided into fixed and price-elastic components, with the price-elastic portion permitted to be fully curtailed. This assumption may overestimate demand flexibility, as many real-world demand response programmes shift or reduce consumption rather than eliminate it entirely. [9] adopts similar methodology for modelling DR however instead uses DR to reduce the curtailment of variable renewable energy although taking a deterministic approach. DR is modelled using a time-of-use pricing scheme, where changes in electricity demand are determined through the price elasticity of demand. The shifted demand is associated with a marginal cost that is included in the objective function, which seeks to maximise system profit. Unlike the previous study, however, the model imposes a zero-net-energy constraint on demand shifting, ensuring that demand is shifted not curtailed. This approach is also used by [10] modelling DR as a continuous, elastic second-stage resource that responds to generation fixed in the first stage.

2.2 Mix of Flexible and Fixed DR Resources

Another set of papers model DR as a set of both Fixed and Flexible resources, through the use of stochastic two stage optimisation. [18] formulates micro-grid operation explicitly across a day-ahead and a real-time stage, solving the DA and RT stages jointly as a single one-shot two-stage stochastic optimisation, in which demand response resources are split into fast-responsive demand and slow responsive demand. Slow-responsive DR is committed in the DA stage, while fast-responsive DR instead receives a scenario-dependent RT schedule, conditional on the DA-stage commitments already made for the slow-responsive resources. Similar to the previous papers mentioned both DR resources are modelled using a continuous shift-up/shift-down variables that are bounded by fractions of demand, where activation is determined by marginal costs for an additional unit of DR within the objective function, which is different to the price elasticity formulation used in previous papers. Finding the inclusion of Fast demand response smooths the demand profile over time as well as between scenarios, while slow responsive demand has the same schedule across all scenarios, and in conjunction significantly reduce the cost of operating a micro-grid. [11] uses a stochastic two-stage optimisation, where the first stage consists of a network-constrained unit commitment in a base case, while the second stage ensures robustness in security assurance across system scenarios. DR is incorporated as a reserve analogous to up-spinning reserves from generating units, with customers reducing their demand within a predefined lead time, by a fixed amount, to provide the service. The first stage consists of the cost of scheduling the DR reserves, while the second stage includes the cost of deploying the DR reserve within each scenario. DR resources are modelled in the form of DR Provider participation as a discrete bid-quantity offer submitted to the Independent System Operator (ISO), where individual retail customer responses are aggregated into blocks rather than allowing a continuously adjustable shift in demand. Each DR reserve offer is represented as a set of discrete quantity levels, each with an associated cost. From which a mixed-integer formulation selects which blocks to activate, with the minimum activated quantity bounded by the DR program's minimum curtailment level. The offered blocks are constrained to be monotonically increasing in cost, to reflect the assumption that demand curtailment becomes more expensive as the quantity curtailed increases. A DR provider submits two separate offers, a capacity cost paid regardless of deployment and an energy cost paid only if the reserve is actually deployed by the ISO, mirroring the capacity-versus-energy payment structure of conventional spinning reserve markets [11]. This is a different activation mechanism to [18], who model demand response as a continuously adjustable shift-up/shift-down power variable bounded by a fraction of demand and costed directly in the objective, rather than as a discrete menu of pre-priced blocks selected via binary variables. While this doesn't use the idea of fixed and flexible DR resources, it does use a similar underlying idea, where second-stage DR decisions are constrained by commitments made in the first stage. These DR decisions are therefore not just

a recourse variable added to provide flexibility within the model, but a structured response shaped by those earlier commitments. A similar block-activation structure appears in aggregator scheduling models [28], where reserve capacity is constrained by a minimum offer size and a minimum delivery duration, further supporting a discrete rather than continuous view of flexible DR activation. All of the aforementioned papers take solely a cost-minimisation or profit-maximisation approach, which may lead to undesirable ramps within the net load profile that may not be feasible in real-world applications. Furthermore, by using either these fixed reduction blocks or a continuous variable to represent load curtailment, this entirely ignores the rebound effect, where DR resources may attempt to recover some of the lost load during the interruptible period. Leading to an unrealistic load curve.

2.3 Peak and Ramp Aware Formulations

In addition to the typical cost minimisation formulation, few papers do take a ramp and load peak aware formulation. [12] proposes a two stage stochastic model, which DR is again used as a flexible resource, however includes flexible ramping costs, which are used to minimise the undesirable ramping within the net load. Finding that incorporation of DR flattens the load profile globally not just at the peak. However, similar to previous papers, [12] models DR as a continuous variable within the model, which may give the model access to too much flexibility. A related approach [27] constrains the net-load ramping rate directly, however instead of using DR, battery energy storage system (BESS) is used as the flexible resource. The authors do this by formulating a day-ahead BESS scheduling problem for load levelling, where they penalise violations of a net-load ramp-rate limit within the objective function rather than purely a cost minimisation approach. Sensitivity analysis across several ramp-rate limits reveals a counter-intuitive result, a looser ramp constraint produces a flatter net load profile overall, since the net load is then able to change more quickly to reach the target level. However, when the ramp-rate limit is too restrictive, the model cannot satisfy it and the model has to accept the associated penalty, so the net load will still violate the constraint. This suggests that enforcing a tight ramp constraints is not just a trade-off between the flatness of the profile and the cost for doing so, however an overaggressive limit can fail to constrain the profile at all and become self-defeating. This reinforces the case made by [12] that ramp-awareness needs to be built into the formulation carefully rather than imposed as an arbitrary bound. Instead of minimising ramps, [13] takes the approach of minimising the peak to average ratio as part of a multi objective formulation which considers costs and dissatisfaction. Suggesting that using naive peak minimisation just shifts the peak to valley periods and as such creates additional issues, therefore also constrain against the creation of new peaks. This paper also models DR based on price elasticity and demand shifting with a continuous variable for the load reduction, this may prove unrealistic and provide excess flexibility within the model, which allows for these new peaks to arise. While not directly peak minimisation, [25] recognises the use of time-of-use programs may lead to the peak load shift, where the peak is not reduced but shift by common consumption behaviour among customers.

[26] instead adopts a predict-then-optimize framework, using probabilistic time-series forecasting to generate load scenarios that feed a two-stage stochastic BESS scheduling model. Finding peak reductions of up to 26% with further gains as forecast accuracy improves. The objective function consists of operational cost, maximum peak, as well as penalty terms to consider operational preferences. The peak is represented by the maximum per-period electricity taken from the grid. Within this formulation minimising total costs and the peak is not a conflicting objective given shaving peak load reduces the cost of electricity consumption during peak period which has a substantially higher cost.

2.4 Direct Load Control Contracts

Another form of DR is through Direct Load Control Contracts (DLCC's), which may be more representative of how a utility would implement a demand response program in comparison to the continuous curtailable load used in the previous papers. A DLCC gives the utility the right to interrupt or reduce the load for a enrolled device. These calls are subject to pre-negotiated limits on the frequency and duration of the calls. As such these do not allow for continuous adjustment of the load, instead gives

the utility a discrete decision, where a contract can either be called or not. Much of this literature is focused on specific load types for example [14] looks at a specific use case of direct load control in which water heaters could automatically be disconnected during peak periods of the day, highlighting an average reduction in consumption of 0.5kWh/h during the disconnections, as well as a rebound in consumption of 0.28 kWh/h per household. The authors suggest these rebounds can cause new peaks if not controlled as coordinated rebounds between many devices can occur. This introduces the idea of a fixed response profile, where a load’s initial reduction and subsequent rebound follow a predictable trajectory across multiple devices. Water heaters, for example, can be shut off until inbuilt temperature constraints are reached, then follows a fixed, predictable recovery in order to restore the temperature to the correct level. However, the authors do not incorporate this into an optimisation model for scheduling nor incorporate these DLCCs into an operational problem. Group-based extensions of direct load control, such as the method proposed by [21], address the operational challenge of coordinating control across many devices specifically air conditioning simultaneously. Where they must stagger or cap group-level activations to avoid synchronized rebound effects when multiple loads return to service at once.

2.5 Scheduling DLCCs

The incorporation of DLCCs into optimization models is limited, with existing work typically focusing on scheduling over longer horizons. For example, [16] studies DLCC scheduling in which each call has a variable duration and starting time, subject to annual limits on the total number of calls and total hours a resource may be called. They show that the deterministic version of this scheduling problem is NP-complete, and develop an aggregation-disaggregation heuristic for implementing DLCCs that is asymptotically optimal as the length of the contract horizon grows. [15] instead studies DLCC scheduling under a threshold policy, in which a utility can interrupt a callable resource in order to bring load down to a target threshold. This subject to constraints on the number of calls permitted per day and per month and conditional on this target being achievable. The purpose of the threshold policy is to flatten the load curve by shaving peak demand. Furthermore utilities are also required to communicate a target plan to suppliers in advance, so the schedule must be committed prior to the interruption. The full multi-period problem decomposes into a sequence of daily sub-problems, each of which solves for the minimum number of calls needed that day to bring the load down to the threshold. Unlike [16], who allow calls of variable duration, [15] model each call as reducing the load by exactly one unit, so the number of calls maps directly onto the amount of demand reduction achieved. This daily scheduling problem is also NP-hard, and they develop a heuristic together with approximation methods to solve it at scale. Both of these papers work at lower temporal granularities, where the fine details that matters within ramp events for realistic operating scenarios is lost. Increasing the granularity results in a trade off between computational tractability. Where finer time steps significantly increase problem size and reduce the number of scenarios that can be tractably solved, which raises the question of how many scenarios are actually needed to adequately capture load-forecast uncertainty. [16] further characterize the error introduced by their approximation. The relative information error from the deterministic approximation is $O(\frac{1}{\sqrt{n}})$ and the aggregation/disaggregation error is $O(\frac{1}{n})$, where n denotes the length of the horizon; the total relative error is therefore $O(\frac{1}{\sqrt{n}})$. This confirms the idea that using longer scheduling horizons gives a better approximation, a trade-off similar to the one explored next for considering temporal resolution.

2.6 Temporal Resolution in Unit Commitment and DR Scheduling

All the papers mentioned previously operate at the hourly resolution, the hourly resolution can lead to sub optimal solutions given the ramping is largely ignored, as small ramping events that can threaten grid security are hidden between these hourly loads. The impact of granularity in time steps has been extensively studied within the Unit Commitment (UC) literature, however has not been studied within the demand response literature. While the majority of papers addressing operational or UC problems do so at 60-minute resolutions, typically done to address computational cost. [22] examines the impact of time resolution on the efficiency of the UC solution, where using a 15-min resolution

resulted in cost reductions due to improved decision making despite the additional expense of increasing processing requirements. [23] looks at the UC problems using time steps of 5, 15, 30 and 60 min, in which they find higher resolutions can better account for a greater range in fluctuations in system load. As such higher resolutions capture significant ramping that cannot be recorded with resolutions that are not as fine. This is further supported by [24], who show that ramp constraints determine how quickly units can enter operation, so capturing them accurately affects both the objective value and the resulting commitment schedule. Clearly, increasing the granularity has significant trade-offs in that it significantly increases computational complexity and reduces model scalability, which limits the number of scenarios that can be used, raising the question of how many scenarios are needed to sufficiently capture the uncertainty associated with load prediction. This is not a universal limitation, however some DA/RT stochastic micro-grid formulations have already moved to sub-hourly RT resolution specifically to preserve this level of accuracy. [29] adopts an hourly day-ahead stage nested with a 5-minute real-time dispatch stage, explicitly motivating the finer RT resolution as necessary to achieve an accurate and optimal economic formulation, rather than treating hourly resolution as sufficient throughout. This suggests that the coarser resolution used in the DLCC scheduling literature discussed above is not a fundamental modelling limit, but a choice that finer-grained DA/RT micro-grid formulations elsewhere in the literature have already moved past.

2.7 Growing Uncertainty from AI Data Centres

Conventional electricity demand has historically been dominated by predictable consumer and industrial behaviour, producing a daily peak whose timing can be forecast with high confidence. The rapid growth in load from AI Data centres is starting to erode this assumption. Data centres are projected to account for an increasing share of total grid demand, with U.S. estimates rising from roughly 4% today to between 9% and 17% by 2030 [1, 2]. This may significantly impact power system reliability, given their load behaviour is not fully understood. Unlike conventional load, AI data centre demand is tied to workload scheduling rather than predictable human activity, and exhibits both fast and large magnitude fluctuations that are harder to forecast [3]. NERC’s Large Loads Task Force suggests these loads are capable of producing reliability impacts that are comparable to conventional generation-loss contingencies, due both the magnitude and the ramp-rate uncertainty of these loads [5]. This impact is quantified by [31], stating large data centres can consume tens to hundreds of MW of power, with ramping of several MW of power in a few seconds. Therefore as the share of load contributed by AI data centres grows, their own load pattern may shift when the system peak occurs rather than purely its magnitude and introduce unexpected ramps into the profile. [4] frames the uncertainty as acting jointly on both the magnitude and timing of load surges.

2.8 Benefit of Stochastic Approaches

As seen in the literature, stochastic approaches are widely adopted for demand response and micro-grid scheduling, with many studies reporting improvements relative to alternative approaches. However, relatively few explicitly quantify the value of incorporating stochasticity by comparing the stochastic solution with an equivalent deterministic formulation. This distinction is important as demonstrating that a stochastic model performs better does not directly quantify the value gained from accounting for uncertainty. [32] directly evaluates the value of incorporating stochasticity in a micro-grid with DR. DR is modelled as a continuous load-shifting resource, where demand can be increased or decreased within a specified percentage of the original load profile while maintaining the same total daily energy consumption. This allows the model to arbitrarily reshape the load within the specified bounds, providing a high degree of flexibility. DR is treated as a second-stage decision, allowing the load profile to adapt to the realised scenario. The authors also evaluate the Value of the Stochastic Solution (VSS) by comparing the stochastic formulation with a solution based on expected values. Although the calculated VSS is relatively small at 1.9%, the authors identify it as significant, demonstrating that incorporating uncertainty can provide value even when the resulting improvement in the objective is relatively small. Similarly, [33] incorporates DR into a day-ahead micro-grid bidding problem by modelling both time-of-use and emergency DR. Flexible load is represented through shiftable

demand and load elasticity, allowing demand to respond to electricity prices and renewable generation outages. The authors formulate the problem as a stochastic mixed-integer non-linear programme, incorporating uncertainty in load, renewable generation and renewable-unit outages. The VSS is then used to quantify the benefit of incorporating stochasticity into the bidding problem. The reported VSS is 296 compared with an objective value of 8,573, representing an improvement of approximately 3.5%. Although this represents a relatively modest improvement, the authors identify the value of incorporating stochasticity as significant. However, as with [32], the study does not examine how the flexibility available within the DR resource affects the value of stochasticity. While not specifically considering DR, [34] provides an indication that the value of stochasticity may depend on the flexibility available to the system. The study considers alternative forms of flexibility through battery storage and finds that when sufficient real-time flexibility is available from storage, the VSS is zero.

This suggests that the value of incorporating stochasticity is not only dependent on level of uncertainty present, but also by the ability of the system to respond to that uncertainty. Where sufficient flexibility is available, the impact of uncertainty on the resulting decisions may be reduced, diminishing the value of stochastic optimisation. Therefore, these studies demonstrate that incorporating stochasticity can provide value within micro-grid scheduling including in problems that incorporate DR. However, this value is not explicitly investigated within a pure DR scheduling problem, under differing sources of uncertainty. This motivates the first contribution of this dissertation, which is to quantify the value of incorporating stochasticity within a pure DR scheduling problem.

2.9 Flexibility within Demand Response Contracts

While the value of DR as a source of flexibility considered extensively within the literature, far less attention is given to the value that flexibility within the DR resource itself can provide. The flexibility of a DR resource is not determined solely by the magnitude of its interruptible load, but also by other contractual restrictions such as when a resource can be activated, the duration of the activation and the number of calls that are permitted. This is particularly relevant for DLCCs, where the utility can directly control loads subject to predefined contractual limits. [16] models these restrictions through limits on the number and duration of calls, while [36] distinguishes between scheduled and conditional DR contracts, demonstrating how contractual conditions determine when flexibility is available. Similarly, [35] explicitly incorporates the time interval in which a controllable load can be used as part of the contract design. These studies demonstrate that contractual conditions determine how a DR resource can be used within an operational problem. However, they treat these contractual restrictions as fixed when evaluating the incorporation of DR contracts, rather than quantifying the benefit of increasing the flexibility available to an existing DR resource. This motivates the second contribution of this dissertation, which evaluates how changes in the flexibility of DR contracts affect the resulting scheduling solution.

3 Methodology

3.1 Modelling Choices

This project has two objectives, the first concerns value of incorporating stochasticity into the scheduling problem relative to deterministic formulations based on an expected-value forecast. The second concerns evaluating the value in flexibility within a DR contract. The modelling choices below are organised around these two objectives.

3.1.1 Choice of Objective Function

Rather than modelling market prices, generation costs, and DR costs explicitly, this dissertation assumes that the utility is allocated a fixed daily demand response budget together with a predefined portfolio of DR resources that may be called in both the day-ahead and real-time stages. This assumption isolates the scheduling problem from other sources of uncertainty, such as electricity price volatility, allowing the effects of load uncertainty and contract flexibility on scheduling decisions to be examined independently.

Subject to this fixed budget, the utility must determine which DR resources to activate and the start time of each interruptible period, in order to minimise both the peak demand and the overall ramping of the net load profile. A smoother load profile enables generation resources to be scheduled more efficiently and reduces the operational challenges associated with meeting short-term changes in demand.

3.1.2 Uncertainty in Scheduling

Previous studies discussed in section 2 that addresses uncertainty do so through a two-stage stochastic programming model, while few use a multistage model. These are typically computationally challenging even at lower resolution such as hourly, and therefore may rely on heuristics to obtain a feasible solution. One of the main limitations of a two-stage approach for this application is that the second-stage decisions would be made based on the entire realised load profile. In practice, this information is not available when scheduling decisions are made. Instead, decisions are updated as new forecasts become available throughout the day. The multistage formulation adopted here addresses this issue by allowing real-time demand response resources to be scheduled at noon using a partially resolved forecast rather than the fully realised load profile, thereby providing a more realistic representation of the information available to the utility.

Beyond correcting this information leakage, the multistage formulation enables both the value of stochastic optimisation and the additional value of staged recourse to be quantified. Fixed DA resources are committed once against the day-ahead forecast, while flexibility is provided by RT flexible resources which are committed against a partially resolved load profile at noon. This benefit is quantified using the value of the stochastic solution (VSS) and the expected value of perfect information (EVPI). This allows for the separation of the benefit of stochastic information at the first stage and the benefit of flexible recourse at later stages. The size of this benefit will depend on the degree of variability in the underlying load profiles. This is tested using a representative winter-peak profile as a baseline case and higher-variability conditions, such as those introduced by AI data centre driven demand as well as uncertainty in the net load from wind generation.

3.1.3 Demand Response with Fixed Response Profiles

As discussed in Section 2, the majority of the literature models DR as a continuous, price-elastic resource that provides flexibility through a recourse variable [7, 9]. A recurring limitation identified in the literature is that continuous formulations allow demand to be curtailed by an arbitrary amount at each time step [7, 13]. This implicitly assumes that the utility has full flexibility in how much demand to shift or curtail, without significant restrictions, which can overstate the flexibility available from real demand response resources.

Continuous formulations may also produce unrealistic operating schedules. As shown by [13], the optimiser can create new peaks or valleys in the load profile rather than simply reducing the system peak. In practice, utility demand response programmes are rarely implemented in this manner. Instead, they are commonly based on Direct Load Control Contracts (DLCCs), which give the utility the right to interrupt an enrolled device subject to pre-negotiated limits on call frequency and duration [14]. Under these contracts, the response of a resource is a discrete, bounded action rather than a continuously adjustable quantity.

The formulation adopted in this dissertation therefore follows the DLCC framework rather than the continuous-elasticity framework. However, unlike the DLCC scheduling models reviewed in Section 2, each resource is represented by a fixed load profile that is either activated in its entirety or not activated at all. The utility’s flexibility is therefore limited to selecting the start time and, where applicable, the duration of the interruptible period, while the warm-up and rebound behaviour before and after the interruption is predetermined. To maintain computational tractability, these combinations of start times and interruptible durations are represented as a set of enumerated fixed load profiles with a constant reduction magnitude. This approach provides a more realistic representation of how demand response contracts are exercised in practice while also allowing the effect of contractual flexibility on scheduling performance to be investigated.

3.2 Source of Load Data

Hourly load data is sourced from PJM, specifically the Atlantic City Electric Company coverage area. A representative peak demand day is selected, the 10th of December 2025, chosen for its pronounced winter peak demand spike, providing a suitable baseline scenario for this study. Given the requirement for the project to be open-sourced, only publicly available data could be used, however, public historical load data from utilities is generally not available at sub-hourly resolution, so no winter-peak profiles at the 5-minute resolution was available. The hourly series is therefore interpolated to 5-minute granularity, after which synthetic noise is added to recover the short-term ramping characteristics of higher resolution load profiles. The magnitude of this noise directly determines the scenario variance available to the stochastic formulation, and therefore affects the value of a stochastic formulation and is discussed in Section 4.1 alongside the resulting VSS and EVPI results.

3.3 Scenario Generation

The multistage scenario tree is constructed using the Fast Forward Selection (FFS) algorithm proposed by [17]. The algorithm is applied to an initial scenario fan generated using the approach described in [18], in which uncertainty is introduced around a deterministic load forecast. This methodology models load uncertainty by adding independent Gaussian perturbations to each time period. Although this approach is adequate for hourly resolution, it becomes less realistic when the scheduling horizon is represented at 15-minute or 5-minute intervals. This is due to the assumed independence of the perturbations, resulting in significant fluctuations between subsequent times steps that are inconsistent with the behaviour of a real load profile. In practice, forecast errors tend to persist over multiple intervals, resulting in a smoother evolution of demand over time.

To capture this temporal dependence, forecast errors are modelled as a first-order autoregressive (AR(1)) process:

$$\epsilon_t = \rho\epsilon_{t-1} + \eta_t,$$

where ϵ_t is the forecast error at time interval t , ρ is the autoregressive coefficient, and η_t is an independent normally distributed random variable. The parameter ρ determines the persistence of forecast errors, with larger values producing more strongly correlated deviations across consecutive time intervals.

The load for candidate scenario i is then obtained as

$$L_{ti} = \hat{L}_t + \epsilon_{ti},$$

where $\hat{L}_t$ is the deterministic forecast load and L_{ti} is the corresponding candidate scenario value. This formulation preserves the underlying forecast trajectory while generating uncertainty that evolves smoothly through time. Temporal correlation is particularly important for the 5-minute model, where without it, independently generated errors would produce unrealistically volatile load trajectories and exaggerated short-term ramps.

Since demand response is primarily intended to mitigate periods of exceptionally high system demand, the deterministic forecast is selected to represent a high-demand day on which demand response is most likely to be activated.

A standard deviation equal to 2% of the forecast load is used to generate the initial scenario fan. As the magnitude of uncertainty directly influences the potential value of stochastic optimisation, a sensitivity analysis of the scenario variance is presented in Section 4.1. This analysis examines how the assumed level of uncertainty affects the Value of the Stochastic Solution (VSS) and the Expected Value of Perfect Information (EVPI).

Due to computational limits 3000 of these candidate scenarios are generated using this approach, from which the multistage scenario tree is constructed. The 24-hour horizon is divided into two 12-hour stages. FFS is first applied to the initial scenario fan over the first stage to identify a fixed number of representative noon nodes. Each original scenario is then assigned to the nearest representative node according to its first-stage load trajectory. FFS is then applied within each resulting cluster to select representative second-stage trajectories. The final scenario tree therefore consists of a set of representative morning paths, each of which branches into a collection of conditional afternoon paths. This preserves the sequential structure of uncertainty while substantially reducing the number of scenarios that must be considered in the optimisation model.

3.3.1 Scenario-Dependent Peak Timing

Modelling the uncertainty as perturbations around a fixed load profile results in the time index at which the peak occurs being identical across all scenarios by construction. Between scenarios, only the magnitude of the peak and the variability of the ramp around it differ. This may have significant impacts for the DR scheduling problem studied here. Under the noise type uncertainty currently used, since the peak occurs at the same time index in every scenario, variation in peak magnitude across scenarios has limited effect on which DR resources are called. The model can just stack DA and RT resources at the same time in every scenario to shave the peak. Instead the main effect of this uncertainty is within the ramping, where additional RT resources may be required to control the resulting ramp profile even though the peak reduction is unaffected. Introducing this peak timing uncertainty changes the problem. The model can no longer commit to a single fixed window and must instead hedge DR activation across the range of times at which the peak may occur, which is closer to the risk actually posed by increasing AI data centre penetration and motivates extending the scenario generation process beyond perturbations around a fixed profile.

To capture this peak timing uncertainty, each of the n_{initial} trajectories is generated by applying an independent time warp to the historical load profile. A peak-shift offset $\delta_i \sim \mathcal{N}(0, \sigma_{\text{peak}}^2)$ is drawn for each candidate scenario i , giving a perturbed peak location $\tilde{t}_i^{\text{peak}} = \text{clip}(t^{\text{peak}} + \delta_i, 1, T - 2)$. A piecewise-linear warp is then constructed from three control points, the start of the horizon, the (shifted) peak, and the end of the horizon, mapping each output time step to a source time in the original profile through linear interpolation. The scenario trajectory is obtained by interpolating the historical load at these warped source times, preserving the overall shape of the daily profile while advancing or delaying the peak.

Once this new trajectory is found, the same scenario reduction process is used as found in subsection 3.3 using the AR(1) process to add noise, then using FFS to generate the representative scenario tree.

3.3.2 AI Ramping Events

To simulate the impact of significant ramping events, additional spikes are generated from a poison arrival process which is superimposed on the 3000 scenarios independently prior to the scenario

reductions process.

For each trajectory, the number of ramping events is drawn $n \sim \text{Poisson}(\lambda)$, with each event's onset time sampled uniformly between noon and the end period. Each of these ramping events are modelled as a short ramp, a hold period of random duration between 1 and 4 periods, and a symmetric ramp-down. The magnitude of this ramp is drawn as a signed fraction of the base load in the candidate scenario i at the onset time, $\text{mag} = s \cdot |\mathcal{N}(\mu_{\text{frac}}, \sigma_{\text{frac}}^2)| \cdot L_{ti}$, where the sign s is positive with probability p_{up} . Because spikes are realised independently per scenario, they contribute additional variance between scenarios within the scenario fan rather than just widening dispersion around a single expected peak. An example of the resulting scenario tree can be seen in sub-figure b of 1, where distinct ramping events can clearly be seen.

3.3.3 Net Load from Wind Generation

Additional uncertainty is introduced by looking at net load, where wind generation is subtracted from each candidate load scenario to create the net load. $L_{ti}^{\text{net}} = L_{ti} - W_{ti}$, superimposed on the 3000 candidate scenarios prior to scenario reduction. The deterministic wind forecast is taken from historical hourly wind generation published by PJM for the same region and target day as the load, interpolated onto the model time grid, and scaled so that its mean output corresponds to a target wind penetration of 50% of mean forecast load. This reflects an installed wind capacity sized to offset roughly half of expected system demand.

Uncertainty around this scaled forecast is generated using the same AR(1) process applied to the load, with the innovation standard deviation at each time step scaled to 10% of the local forecast wind output rather than a single fixed value, reflecting the comparatively higher relative variability of wind generation at low output levels. Each load scenario is paired with an independently sampled wind scenario, so load and wind uncertainty are treated as independent sources of variation with no explicit weather-driven correlation between them. The resulting net load trajectories replace load as the input to the scenario reduction process described in Section 3.3 to construct the scenario tree. Representative scenarios can be seen in sub-figure c of figure 1

3.3.4 Summary of Uncertainty Cases

The mechanisms described above are combined to construct the four uncertainty cases evaluated in Section 4.

- **Base Case** - Uncertainty is limited to the AR(1) demand perturbations described in Section 3.3. This represents ordinary day-ahead demand forecast error.
- **AI Spike** The AR(1) demand perturbations are combined with the Poisson-arrival ramping events described in Section 3.3.2, representing the additional uncertainty introduced by AI data centre driven demand spikes, however there is no additional uncertainty from peak shifting.
- **Spike + Peak Shift** The AI spike mechanism is combined with the scenario-dependent peak timing described in Section 3.3.1, so both the magnitude and the timing of the daily peak vary across scenarios.
- **Net Load (Load - Wind)** The AR(1) demand perturbations are instead combined with the wind generation described in Section 3.3.3, representing supply-side rather than demand-side uncertainty. This case does not include the AI spike or peak-shift mechanisms.

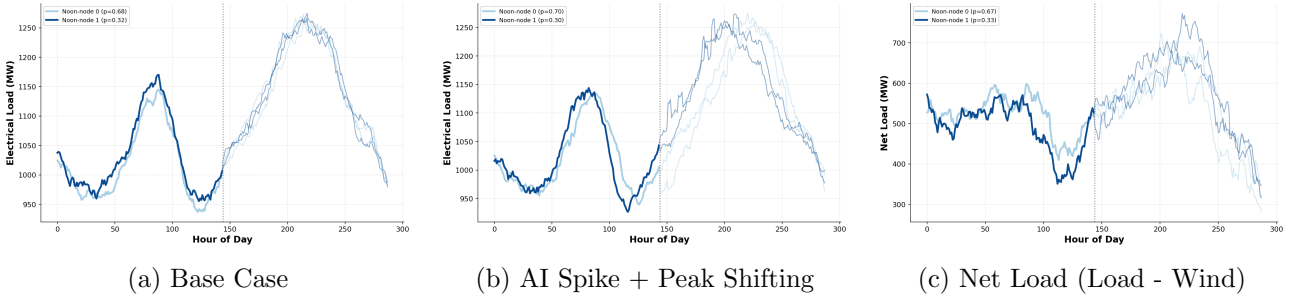


Figure 1: Comparison of scenario trees under different uncertainty representations. Net load scenarios are plotted on a different magnitude scale to the other two panels, since wind generation reduces the net load well below the pure demand profiles shown alongside it.

3.4 Generation of Demand Response Profiles

There are significant delivery regulations within the reserve markets, these define the shape of possible DR profiles. For example, the maximum magnitude for an interruptible load must be reached within a certain time and must be sustained for a given duration [30]. An example of this is within the Italian ancillary service market, where tertiary spinning reserve should be reached within 15 mins and should last for at least 120 min, While for the tertiary replacement reserve should be reached within 120 min and last for 480 min. Furthermore, the aggregators bid/offer size must be a minimum 1 MW. [28]. These regulatory timing and duration requirements motivate the range of contract flexibility parameters, including start-time granularity and activation windows and the size of the interruptible period, which is examined in Section 4.2.

Fixed Versus Real-time Resources

Similar to [18], this paper adopts a two tier structure where fixed DR resources must be scheduled in the day ahead stage, these consist of DR resources such as industrial loads, water pumping, batch industrial processes. As well as flexible Resources that can be scheduled at noon on the same day in order to provide scenario dependent flexibility. These flexible resources include residential thermostatically controlled loads, EV charging, Refrigeration and cold storage, smart appliances. This reflects how a utility in practice may commit DR resources, where some resources must be committed day ahead against a predictable load curve, while others are held in reserve to provide additional flexibility in case additional intervention is needed. As such, DA fixed resources are scheduled once against the day-ahead forecast, and RT flexible resources are committed at the noon-nodes once an updated, partially-resolved forecast is available, giving the model access to corrective flexibility exactly at the point where uncertainty has been partially reduced.

Shape of the Response Profiles

Day-ahead and real-time DR resource archetypes are constructed, these are not intended to represent a particular appliance, but representative classes of DR behaviour. These archetypes consists of a warm-up load for some resources, an interruptible period with a fixed reduction magnitude for a fixed length, followed by an exponential-lag rebound governed by a resource-specific time constant τ . This is highlighted by [36] which acknowledge a load reduction is followed by a load increase to return to a more steady state load. They model this using a two blocks a response block and rebound block, in which the blocks do not identical with respect to the time period and load quantity that is reduced/increased.

This is supported by [14], which measured DLCC of residential water heaters in Norway and found an average reduction of approximately 0.5 kWh/h during disconnection, followed by a pay-back increase of up to 0.28 kWh/h in the hour immediately after reconnection, demonstrating that reconnection produces a bounded, short-lived rebound rather than an unconstrained or instantaneous return to baseline. [20] characterise this behaviour more precisely for electric water heaters, explicitly parametrising flexibility activation in terms of rebound percentage, delay, and ramp rate, and

reporting ramp-up and ramp-down rates as high as 48.5% and 23.8% per minute respectively. This supports representing each archetype’s response as a fixed-shape curve governed by a small number of interpretable parameters, here compressed into a single rebound time constant per archetype, rather than as a free-form continuous trajectory the optimiser is left to choose. As such, five day-ahead archetypes are created, consisting of increasingly long interruptible periods with steps up to a maximum interruptible load magnitude, as shown in Figure 2. There are also eight real-time DR resources, which have a shorter interruptible period and a lower maximum interruptible load. Additional noise is superimposed on the warm-up and rebound segments of each profile, since real load transitions during these periods are not smooth ramps but exhibit irregular, higher-frequency variation. This is done using the same AR(1) process described in Section 3.3.

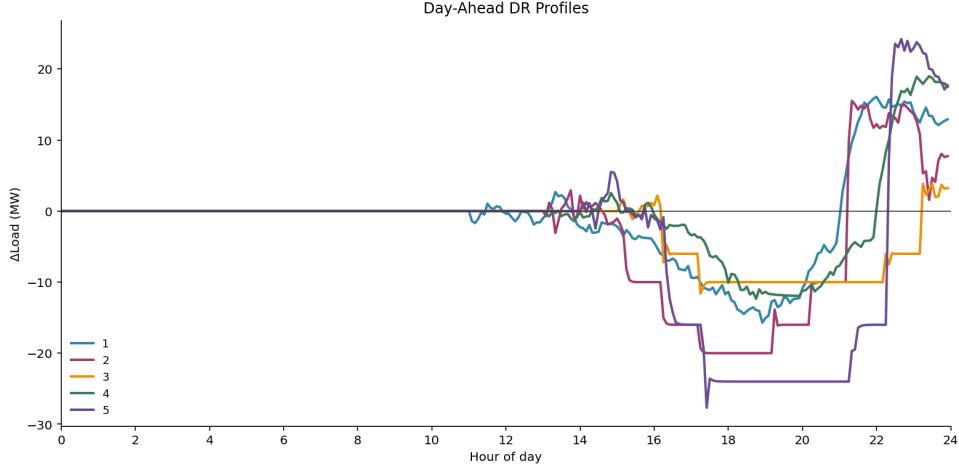


Figure 2: Representative day-ahead DR profiles, one curve per archetype, showing the change in load relative to baseline (MW) at a representative start time. Each archetype curtails for a fixed duration before recovering through an exponential-lag rebound governed by its own time constant τ .

Enumerated Start-time Variants

Representing each flexible DR resource’s timing flexibility as a small enumerated set of start-time variants, rather than as a fully continuous shift decision, follows the same practical logic as the DLCC scheduling literature of [16] and [15], who show that scheduling calls for DLCCs subject to periodic limits is computationally demanding at full generality, and develop aggregation-disaggregation approximations to keep the problem tractable at scale. Enumerating a bounded set of feasible start times per resource here serves an analogous purpose, keeping the RT activation decision within a tractable MILP while still allowing the model to choose, from a realistic and contractually-bounded set, when a given resource’s fixed response profile is best placed, rather than either freezing timing entirely or allowing a fully continuous shift.

This enumerated set is also the mechanism through which the flexibility within DR contracts can be tested. Contract flexibility is measured through the granularity of allowable start times is varied, comparing hourly against 5-minute activation windows. Additional flexibility is provided through varying the length of the interruptible period for these DR resources. This is done to establish how much timing flexibility a contract needs before any additional steps meaningfully improving peak and ramp reduction. The raw profiles before they are enumerated for different possible start times can be seen in Figure 3.

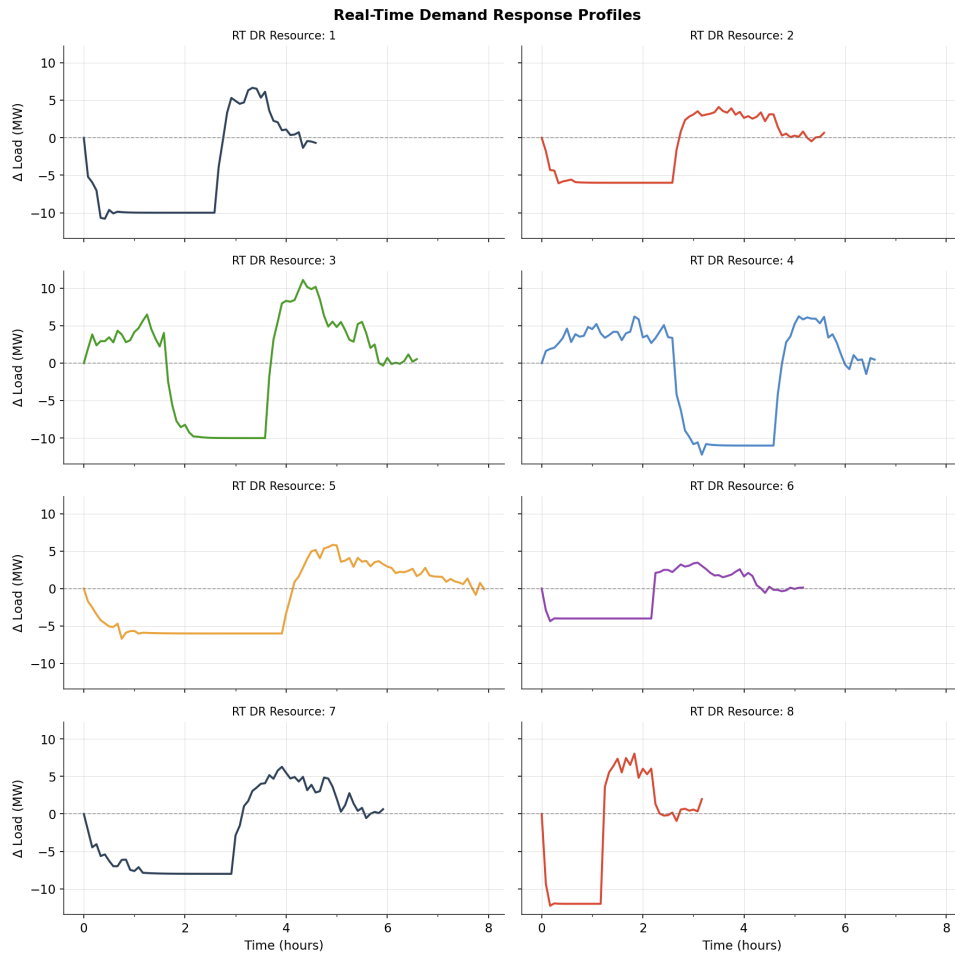


Figure 3: Real-time DR profiles for all eight archetypes, before enumeration across start times, showing the change in load relative to baseline (MW).

3.5 Model

The multi-stage stochastic programming formulation is presented below.

Sets

J	DA fixed DR resources
L	RT DR resources
K	start-time variants DA Resources
I	start-time variants RT Resources
T	time periods
S	scenarios
N	noon nodes

Parameters

L_{ts}	base load at time t in scenario s
δ_{jkt}	response profile of DA fixed DR resource j under start-variant k at time t
ρ_{lit}	zero-padded response profile of RT resource l under start-variant i at time t
DA^{max}	max number of DA fixed DR resources activated
RT^{max}	max number of RT resources activated per node
$rampLimit$	cap on the 3-period rolling absolute ramp, active for $t \geq t_{mid}$
t_{mid}	noon time index from which the rolling ramp constraint applies
$n(s)$	noon node associated with scenario s
λ^{ramp}	penalty weight on ramp-limit breaches

Variables

L_{ts}^{net}	net load at time t in scenario s after DR impact
ΔL_{jt}	load change by DA fixed DR resource j at time t
$\Delta L_{l,t,s}^{RT}$	load change by RT resource l at time t in scenario s
u_{jk}	binary, 1 if DA fixed DR resource j is activated with start-variant k
$u_{l,i,n}^{RT}$	binary, 1 if RT resource l is activated with start-variant i at node n
π_s	auxiliary variable for peak net load in scenario s
r_{ts}	auxiliary variable for absolute ramp between $t - 1$ and t in scenario s
v_{ts}^{ramp}	slack variable for excess ramping beyond limit at time t in scenario s

3.5.1 Objective Function

$$\min \sum_s P_s \pi_s + \sum_s P_s \sum_t r_{ts} + \lambda^{ramp} \sum_s P_s \sum_{t > t_{mid}} v_{ts}^{ramp} \quad (1)$$

The objective in Equation 1 minimises a weighted combination of expected peak net load, expected total ramping, and a penalty for breaching a 3-period rolling ramp limit, all evaluated as expectations under the scenario probabilities P_s with all variables measured in MW. The first term is the expected peak π_s ; the second is the expected sum of absolute period-to-period ramps r_{ts} over the horizon T ; the third penalises breaches of the rolling ramp limit through the slack variable v_{ts}^{ramp} . All three terms depend on the scheduling of DA fixed and RT flexible DR resources, which reshape the net load profile from which π_s and r_{ts} are derived.

The first two terms are motivated in Section 3.1.1, while the benefit of the penalty term needs more justification. A sharp ramp concentrated in a few periods strains the grid beyond what the aggregate ramping term captures, requiring expensive quick-start generation and additional resources to manage the voltage instability caused by these ramps. Therefore these large ramps need to be specifically targeted, not merely as a contributor to the total ramping. The aggregate ramping term alone does not achieve this, since a large local ramp can be offset by smaller ramps elsewhere in the horizon without materially changing the total, leaving the optimiser with no direct incentive to avoid it.

A hard rolling ramp constraint was tried first but was found to be too restrictive, because the constraint is enforced across every scenario, a single difficult scenario forces the ramping limit to be relaxed well beyond what is needed elsewhere to avoid infeasibility, defeating its purpose. Enforcing it instead as a penalised soft constraint, with a large weight $\lambda = 50$ on violations, keeps the limit binding in the common case while allowing scenario-specific breaches at a cost. This weight was chosen via sensitivity analysis. This initial hard-constraint analysis is omitted for brevity.

3.5.2 Constraints

Net Load

$$L_{ts}^{\text{net}} = L_{ts} + \sum_j \Delta L_{jt} + \sum_l \Delta L_{lts}^{RT} \quad \forall t, s \quad (2)$$

Peak Load

$$\pi_s \geq L_{ts}^{\text{net}} \quad \forall t, s \quad (3)$$

Absolute Ramping Between Periods

$$r_{ts} \geq L_{ts}^{\text{net}} - L_{t-1,s}^{\text{net}} \quad \forall t \geq 1, s \quad (4)$$

$$r_{ts} \geq - (L_{ts}^{\text{net}} - L_{t-1,s}^{\text{net}}) \quad \forall t \geq 1, s \quad (5)$$

Rolling Ramp Limit

$$r_{ts} + r_{t-1,s} + r_{t-2,s} \leq \text{rampLimit} + v_{ts}^{\text{ramp}} \quad \forall t \geq t_{\text{mid}}, s \quad (6)$$

DA DR Resources

$$\Delta L_{jt} = \sum_k \delta_{jkt} u_{jk} \quad \forall j, t \quad (7)$$

$$\sum_k u_{jk} \leq 1 \quad \forall j \quad (8)$$

$$\sum_k \sum_j u_{jk} \leq DA^{\text{max}} \quad (9)$$

RT DR Resources

$$\Delta L_{lts}^{RT} = \sum_{i \in I} \rho_{lit} u_{lin(s)}^{RT} \quad \forall l, t, s \quad (10)$$

$$\sum_i u_{lin}^{RT} \leq 1 \quad \forall l, n \quad (11)$$

$$\sum_l \sum_i u_{lin}^{RT} \leq RT^{max} \quad \forall n \quad (12)$$

Domain Constraints

$$\pi_s \geq 0 \quad \forall s \quad (13)$$

$$r_{ts} \geq 0 \quad \forall t, s \quad (14)$$

$$u_{jk} \in \{0, 1\} \quad \forall j, k \quad (15)$$

$$u_{lin}^{RT} \in \{0, 1\} \quad \forall l, i, n \quad (16)$$

$$v_{ts}^{ramp} \geq 0 \quad \forall t \geq t_{mid}, s \quad (17)$$

Constraint 2 defines the net load at time t in scenario s , consisting of the base load L_{ts} plus the contribution from both fixed and flexible DR resources. Constraint 3 defines the scenario peak π_s as an upper bound on the net load over the full horizon. Constraints 4 and 5 linearise the absolute period-to-period ramp, forcing r_{ts} to bound both the positive and negative differences. Constraint 6 caps the sum of ramps over the current period and the preceding two, active only from t_{mid} onward, and includes the slack variable v_{ts}^{ramp} so a breach of the limit incurs a penalty in the objective rather than causing infeasibility. Constraints 7-9 define the impact of the DA resources, giving the load change from activating a day-ahead DR resource, restricting each resource to at most one start-variant, and capping the total number of DA DR resources used. Constraints 10-12 take the same form for the RT resources, enforced at the node level rather than once for all scenarios.

4 Results

All numerical simulations were implemented in Python using Pyomo 6.10.1 and solved with the MILP solver HiGHS 1.15.1, run on a MacBook with an Apple M1 chip and 8GB of RAM, using a MIP gap of 1%. With full model flexibility, both RT and DA resources activated at 5-minute granularity, the model has an average solve time of 600 seconds. This falls substantially once additional restrictions are placed on the enumerated candidate profiles: with RT resources restricted to 15-minute granularity and DA resources to half-hour granularity, the average solve time drops to 77 seconds. Figure 4 shows an example model solution under the base-case uncertainty (4 scenarios), illustrating typical scheduling behaviour within the base case uncertainty before the detailed sensitivity results that follow. Each panel shows one scenario's base load against the load after DR contribution; across all four scenarios, the peak between approximately 16:00 and 20:00 is shaved by a similar amount and at a similar time, despite the differing scenario probabilities and realisations. This consistency in the solution will impact the VSS result reported in Section 4.1, because the optimal day-ahead decision barely changes across scenarios in the base case, the stochastic solution offers little advantage over a deterministic one built from the mean scenario.

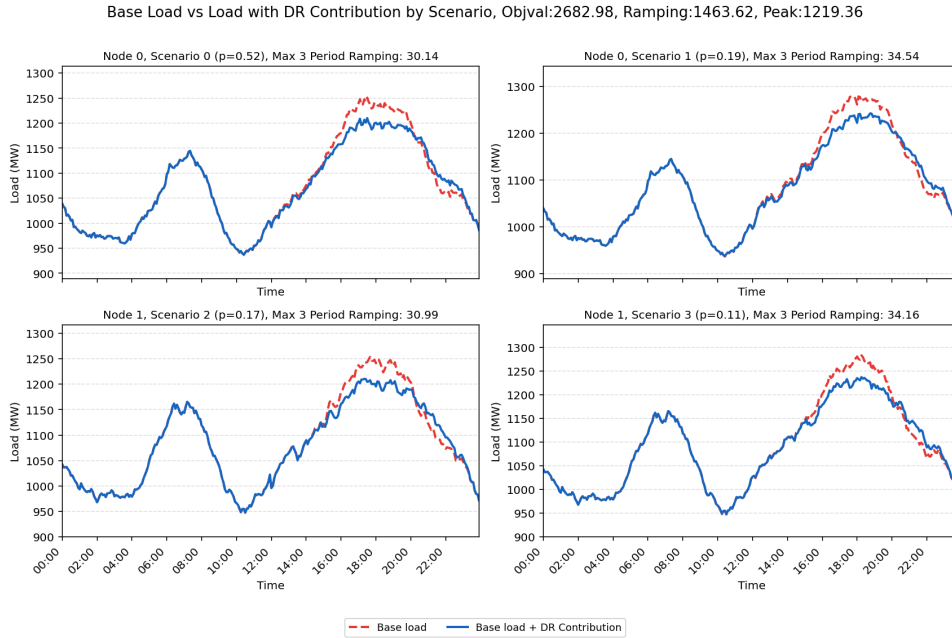


Figure 4: Example day-ahead solution under the base-case uncertainty, showing base load against load after DR contribution for all four scenarios.

4.1 Sensitivity to Uncertainty

	Base Case	AI Spikes	Spikes + Peak Shifting	Net Load (Load - Wind)
VSS	$0.2\% \pm 0.5\%$	$5.8\% \pm 4.6\%$	$7.7\% \pm 4.8\%$	$7.1\% \pm 3.5\%$
VSS t -stat	1.79	5.64	7.17	9.07
EVPI	$1.7\% \pm 0.5\%$	$19.8\% \pm 10.1\%$	$21.0\% \pm 8.5\%$	$10.0\% \pm 4.5\%$
EVPI t -stat	15.20	8.77	11.05	9.94

Table 1: VSS and EVPI averaged across 20 runs (mean $\pm$ standard deviation), percentage is with respect to the objective function. t -statistics test $H_0 : \text{mean} = 0$, critical values are 2.093 (5%) and 2.861 (1%).

Table 1 Shows the VSS and EVPI for the differing uncertainty scenarios. As expected the VSS increases as the uncertainty increases, with the base case having the least uncertainty imbedded while

net-load has the most. The t -statistics show these VSS values are significant and not just sampling noise. In which except for the base case ($t = 1.79$, below the 5% threshold), every VSS value is statistically different from zero at the 1% significance level. The insignificant VSS found within the base case is largely to be expected. This is due to how the load uncertainty is incorporated, where it is perturbations around a fixed load profile. Therefore, peak timing does not vary across scenario, resulting in the load profile within the mean-value problem to share this same peak timing. Since DA resources are used primarily for peak reduction (This is substantiated in subsequent sections), a DA schedule chosen against the load profile in the mean-value problem ends up identical to the one chosen against the full stochastic problem. The two problems therefore select the same DA resources at the same start times, which is why VSS is close to zero here. As uncertainty is added, this breaks down. Peak timing varies across scenarios, so the peak within the mean-value problem no longer matches every individual scenario's peak, and the first-stage DA decision made based on this is no longer optimal for the stochastic problems, therefore increasing VSS. Where there is high variation in ramping such as the AI spike cases, DA resources may also be used to mitigate the spikes themselves, this does not occur in the mean-value problem since spikes between scenarios are smoothed by averaging. Both of these effects widen the gap between the mean-value and stochastic first-stage decisions, which is why VSS grows as uncertainty increases. EVPI is consistently much larger than VSS, which suggests most of the value comes from reacting to the realisations through the Real Time scheduling rather than from better DA commitments. This makes sense given that they are mostly used to attack the ramps which differ significantly between scenarios. [32] and [33] treated VSS values of 1.9% and 3.5% as large enough to justify a stochastic approach over a deterministic one in comparable day-ahead micro-grid bidding problems. It's worth noting these papers address a slightly different problem, day-ahead scheduling and bidding for an entire micro-grid, covering generation dispatch and day-ahead and real-time bidding, rather than solely the task of scheduling demand response resources, so is not a direct comparison. However both are stochastic day-ahead scheduling problems facing comparable sources of uncertainty, and the values found here (5.8-7.7%) are larger than both benchmarks. A gap of this size can account for a significant proportion of operating costs and not just a marginal difference between the two approaches. In the base case, VSS is negligible, meaning a deterministic formulation performs just as well as the stochastic one and should be used instead. This does not diminish the underlying model, where representing demand response resources as a fixed set of enumerated start-time and duration profiles, activated through binary selection variables, is itself a novel formulation of the DR scheduling problem relative to the continuous dispatch and cost-curve representations reviewed in Section 2, independent of whether it is solved deterministically or stochastically. The value of the stochastic extension is conditional on the level of uncertainty present, but the enumerated-profile representation is a contribution in its own right, and remains applicable even in cases where a deterministic solve is sufficient.

Figure 5 shows representative dispatch schedules from three of the uncertainty cases, illustrating how the choice and timing of DR resources shifts as the underlying uncertainty changes. In the base case (sub-figure a), DA and RT resources are stacked together at the peak, RT resources supplement the DA reduction as the DA resources begin transitioning out of curtailment and into rebound, and the rebounds themselves are also stacked, which limits the ramp following the end of the peak period. Sub-figure b shows a similar DA stacking pattern as the base case however shifted forward slightly to reduce the ramping up to the peak, but here RT resources are instead deployed specifically to counter the ramping events that exceed the ramp limit which can be seen at 6pm. Sub-figure c shows a less pronounced peak with less resource stacking overall. Note that the ramp limit needed to be relaxed, from 35 MW to 65 MW, since incorporating wind into the net load profile produced significantly higher underlying ramping than the pure demand load profiles used in the other two cases. Resources here are more evenly spread across the period rather than concentrated at the peak, with both DA and RT resources used to reduce ramping across the whole profile rather than to target a single peak event, this is also due to the fact incorporating wind generation resulted in a less prominent peak.

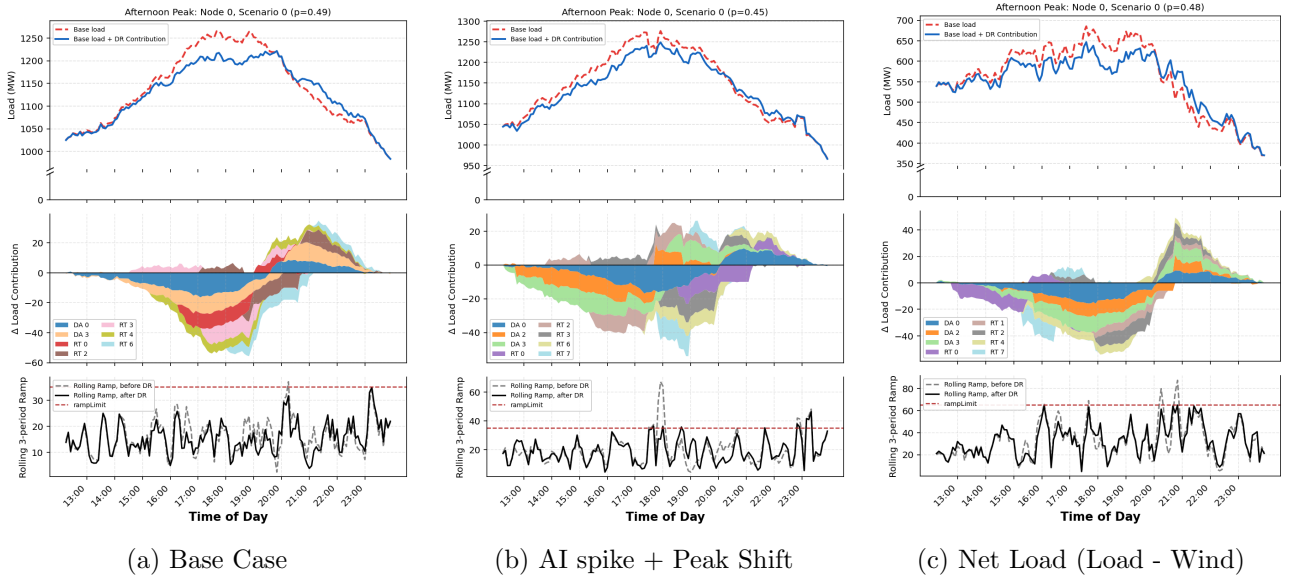


Figure 5: Example dispatch solutions for three uncertainty cases. Each panel shows, top to bottom, the base load against the load after DR contribution, the breakdown of that contribution by individual resource, and the rolling 3-period ramp against the ramp limit. The net load scenario is plotted on a different magnitude scale to the other two panels.

4.2 Impact of Start-time Granularity

4.2.1 Day Ahead Resources

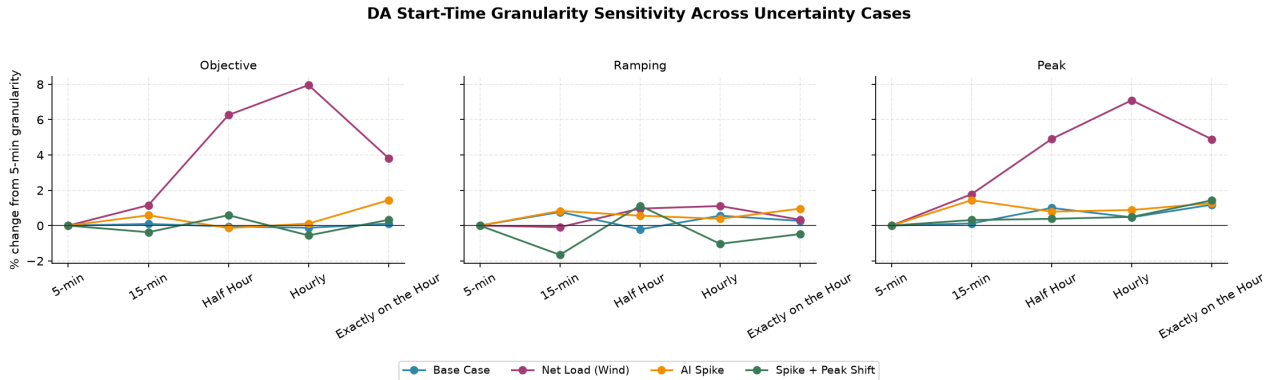


Figure 6: Sensitivity of objective, ramping, and peak to day-ahead start-time granularity, shown as percentage change from unrestricted 5-minute placement.

As seen in figure 6 the objective value is entirely insensitive to changes in the start-time granularity of these day ahead resources in all the base load uncertainty cases with small increase in the peak values of approx 1.5%. This is likely because these DA resources have long curtailment windows, the reduction period runs for around six hours in total (from when the resource first curtails to when it returns to zero, not six hours spent continuously at maximum reduction) so a shift in start time of an hour or less does not change when the reduction can coincide with the peak. Since DA resources are used primarily for peak reduction rather than for targeting ramps, they lack the fine timing precision that would make this type of granularity matter, which is why the objective stays largely insensitive to it. However the net-load case is sensitive to this, with a significant change in the objective of 7.97% for the hourly case, with this coming from both a worsening of the peak value as well as an increase in the violation from the ramping penalty. This gap comes down to a mismatch between how the DA archetypes were designed and what actually drives the uncertainty in net-load within the wind case. In the base, spike, and spike plus shift cases, net load is driven by demand alone. The DA curtailment

profiles were designed on an hourly grid to counter a smooth, roughly hourly-shaped afternoon peak, so locking start times to the hour costs very little. The profile these resources are scheduled against is already hour-aligned.

The wind case is different. The net-load here is demand minus an independently drawn historical wind generation sample, and wind’s ramps and troughs are set by real weather data with no relationship to the clock hour. Additional noise is added to the historical wind generation on top of the demand-side noise already present. The resulting net load signal is smaller in magnitude, since wind subtracts around 50% of mean load. It is also rougher throughout the day, with less defined peaks and more significant ramping, since it now reflects two independent stochastic processes rather than one. Furthermore as DA resources are not just used to target the peak in the net-load case, restricting DA start times to the hour results in a timing offset between resource availability and the optimal response which could be intra-hourly. This is worsened by the less distinct peak and choppy profile, whose critical moments are not hour-aligned. Against that kind of signal, the offset is much larger than it would be against a smooth, demand-only, hourly-shaped peak, which is consistent with the 7.97% objective increase seen here.

While the objective value is largely insensitive to DA start-time granularity within the load cases except net-load (load - wind), a more interesting result can be found within how these DR resources are scheduled. Across every granularity setting within the base case, the same pair of DA resources produces are stacked together as seen in Table 2. These are archetypes 1 and 4, the two slowest-responding resources in the DA portfolio ($\tau = 25$ min and $\tau = 30$ min respectively). No amount of scheduling freedom eliminates this interaction, even with full 5-minute start-time control, the optimizer’s chosen schedule leaves their rebounds overlapping by a similar magnitude to when starts are locked to the hour. This supports the idea that these are used to reduce the peak by stacking multiple DR resources. The Overlap magnitude found in table 2 measures how much two resources’ rebounds coincide in time.

Start-time granularity	Top stacking pair	Overlap magnitude
5-min	DA 1 $\leftrightarrow$ DA 4	222.0
15-min	DA 1 $\leftrightarrow$ DA 4	218.6
Half Hour	DA 1 $\leftrightarrow$ DA 4	214.8
Hourly	DA 1 $\leftrightarrow$ DA 4	234.1
Exactly on the Hour	DA 1 $\leftrightarrow$ DA 4	170.7

Table 2: Dominant rebound-stacking pair across DA start-time granularity settings. DA 1 ($\tau = 25$ min) and DA 4 ($\tau = 30$ min), the two slowest-responding archetypes, remain the largest stacking interaction regardless of scheduling flexibility.

A possible explanation is that both resources rebound windows are wide enough, that no available start-time offset separates them without incurring a larger cost elsewhere. This suggests a preference, the optimizer allows for the the rebound collision because re-timing to avoid it or using another resource to counteract the rebound is not cost-effective. Furthermore stacking similar resources may actually be beneficial by reducing the peak significantly while also reducing ramping by absorbing some the large drop off in post peak load reduction, leading to an overall flatter peak period.

However this explanation isn’t definite. Checking which DA archetypes are actually selected at each granularity level shows archetypes 1 and 4 are the only two ever chosen in the base case, from unrestricted 5-minute placement all the way to starts locked on the hour. Since the possible choices of DA resources never changes, this can’t tell us whether 1 and 4 dominate stacking because they’re slow-responding, or simply because they’re the two resources the optimizer always prefers here regardless of response speed. What it does show is that flexibility doesn’t affect the alignment of resources, keeping this overlap between DA resources, where the overlap magnitude stays in a similar range (170 to 234) across every granularity setting, even with full 5-minute freedom to space the rebounds apart. Testing the response-speed hypothesis properly would mean varying either the DA budge or adding additional DA resources so selection isn’t constant, which is left for future work.

4.2.2 Real Time Resources

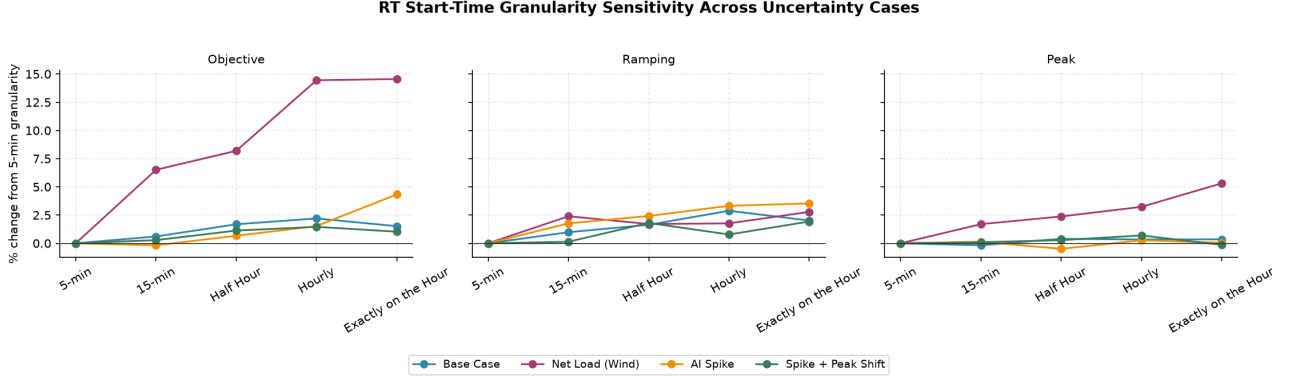


Figure 7: Sensitivity of objective, ramping, and peak to real-time start-time granularity, shown as percentage change from unrestricted 5-minute placement.

As start-time restrictions increase, the base case objective is affected most under the hourly restriction, rising by 2.2%. Almost all of this comes from ramping, which deteriorates by 2.85%, while the peak value increases by only 0.4%. This is a larger effect than it first appears, as in section 4.2.1 the objective is largely insensitive to changes DA start-time granularity, with peak moving by only around 1.5% at most and the objective barely changing at all. RT resources are therefore more sensitive to scheduling granularity than DA resources, which is consistent with the different roles that these resources provide. In which DA resources operate over multi-hour windows where the exact start-time is less important, while RT resources are deployed to counter short, timed ramp events, so losing access to fine grained timing removes their ability to attack these events. This result illustrates the distinct roles of the two resource types: day-ahead resources are scheduled primarily to reduce the peak, while real-time resources owing to their inherent timing flexibility are predominantly responsible for controlling ramping within the load profile. A similar pattern holds across the other uncertainty types, with the majority of the cost accruing from the ramping term. This does not hold for the net load case, where there is significant worsening in the peak value of 5.3%. The objective increases by 14.5%, the majority of which comes from increased violation of the penalty on the 3 period ramping constraint. The peak likely happens due to increased peak time uncertainty between scenarios as such the DA alone cannot be used to reduce the peak time like what is seen in the base case so RT resources must also be used. As with the DA resources, this comes down to timing, where net-load under wind is driven by highly variable wind generation rather than a smooth demand curve, so its ramps have no reason to land on the hour. RT resources are specifically the ones relied on for fine-grained timing control, so restricting their start-times removes exactly the capability needed to track a target whose critical moments are not aligned with the clock hour.

4.3 Length of Interruptible Period

Each DA archetype is given its own set of possible interruptible durations, driven by its response speed and the length of its interruptible period. As the length cap is increased, at each of these lengths resources may choose any of their own durations up to the current length hours, so the optimizer's choice set grows as the length increases. The same approach is used on the real time resources, however the same interruptible durations are shared across all the real time resources. The resulting objective value, expected ramping and expected peak are recorded for each of the different maximum lengths.

4.3.1 Day Ahead Resources

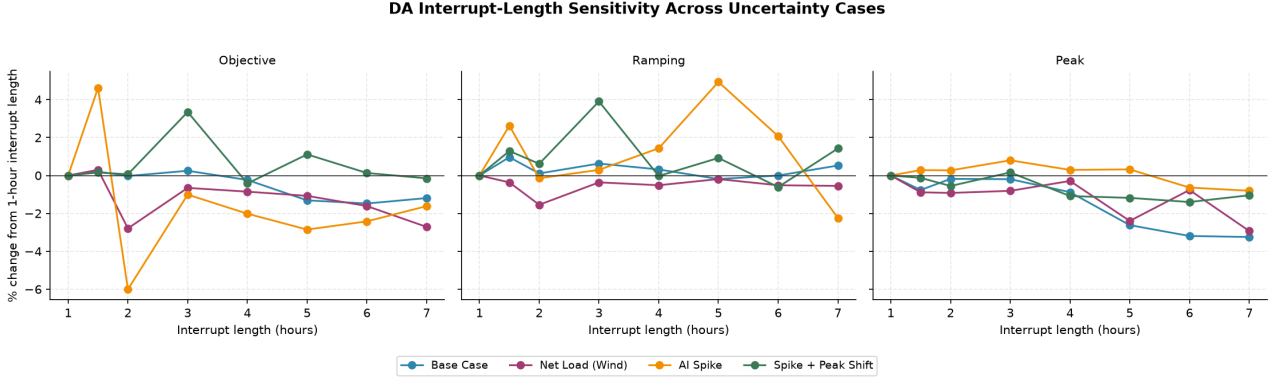


Figure 8: Sensitivity of objective, ramping, and peak to the maximum day-ahead interruptible length, shown as percentage change from the 1-hour length.

Changing the interruptible length has a greater effect than start-time granularity in the majority of uncertainty cases; the exception is the wind uncertainty case, where start-time granularity restrictions affect the peak instead. In the base case, extending the interruptible period from the most restrictive length (1 hour) to the least restrictive (7 hours) reduces the peak by 3.2%, compared to only a 1.1% reduction from relaxing start-time granularity. This is also the largest reduction seen across the uncertainty cases, likely because the base case has no peak-timing uncertainty: with a longer interruptible period, more of the peak can be shaved by stacking DA resources, and their rebounds can also be stacked into the post-peak ramp-down rather than creating a new secondary peak. As a result, RT resources are not needed to clean up these rebounds afterward, an effect visible in sub-figure a of Figure 5. As expected, changes in interruptible length primarily affect the peak value, since in the base case this is primarily the role of the DA resources.

4.3.2 Real Time Resources

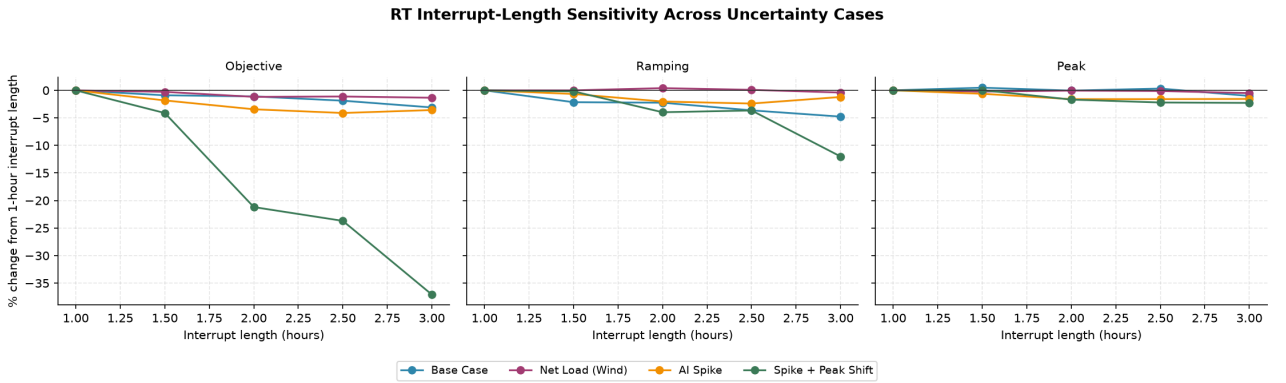


Figure 9: Sensitivity of objective, ramping, and peak to the maximum real-time interruptible length, shown as percentage change from the 1-hour length.

Changes in the interruptible length have significant effect on the objective largely driven by ramping as well as reducing the penalty from violations of the ramping constraint. This can especially be seen in the spike + peak shift case, where reducing the interruptible period reduces its ability to target ramps that violate the ramp limit, which also leads to an increase in ramping across the entire period. While the impact on other scenarios is masked by the spike + shifting case, where the objective drops by 37% while ramping only improves by 12%, this mismatch is from a reduction in the penalty coming from violations of the 3 period rolling ramp limit. There is also a significant reduction in the base case with ramping improving by 4.8% between the most restrictive and least restricting interruptible

lengths. This further highlights the role of these resources, where DA are used primarily for peak reduction and RT resources are used to manage both short term ramping events as well as the overall ramping profile.

5 Limitations

Before stating the conclusions, it is first necessary to understand the limitations of this dissertation. The first is the quality of the load data used, where only hourly load and wind generation data were publicly available, so both were interpolated to 5-minute resolution with noise added to create plausible sub-hourly ramping behaviour. This noise is a reasonable approximation, but it is not a direct measurement of how load or wind actually move at 5-minute resolution, so the specific ramp magnitudes reported here should be interpreted as representative rather than as a precise characterisation of the true sub-hourly profile. Both the load-data and wind generation is from a single historical day, so it reflects one realisation of a singular day rather than possible load conditions.

The DR archetypes themselves are also a limitation. Their shapes were designed to be plausible and were informed by regulatory delivery-timing requirements and published device studies, but they were not specific to any aggregator’s actual product offering. As such, they may not be representative of what a real DR aggregator would offer a utility in practice, and the specific magnitudes and shapes used here are used to illustrate the binary modelling mechanism rather than trying model any particular contract.

6 Conclusion

This dissertation proposed a novel formulation for the DR scheduling problem, where DR resources are represented as a fixed set of enumerated start-time and duration profiles that are activated through binary variables, rather than the typical continuous dispatch and cost-curve representation seen in Section 2. The value of this representation is independent of whether the underlying problem is solved deterministically or stochastically, and provided the basis for investigating two further questions: the value of using a MSP approach over a deterministic one when scheduling DR resources under net-load uncertainty, and whether flexibility in DR contracts, specifically in start-time granularity and interruption duration, provides benefit to the system operator. A MSP model built on this was evaluated across four uncertainty cases, a base case with no peak-timing uncertainty, a case with large AI data-centre driven ramping events, a combined spike and peak shifting case and finally a net load case incorporating wind generation uncertainty.

On the benefit of stochasticity. The value of the stochastic approach was found to depend on the uncertainty present. In the base case, VSS was statistically indistinguishable from zero, therefore there is no value in a stochastic model in which a deterministic model should be used. The underlying reason for this is that within the base case, uncertainty is constructed as perturbations around a fixed load profile, so peak timing does not vary across scenarios. This results in the optimiser choosing the same DA decisions in the mean value problem as the one in the stochastic problem. Despite a lack of value in the base case, the enumerated-profile representation is still interesting even where the stochastic extension adds no value. In every other case, VSS was significant at the 1% level (5.8-7.7%, Section 4.1), exceeding values reported as sufficient justification for using a stochastic approach in comparable micro-grid scheduling literature [32, 33]. Furthermore, EVPI was consistently larger than VSS across all the uncertainty cases, indicating that most of the value from resolving uncertainty comes from real-time recourse rather than from day-ahead commitment. Therefore the choice of a stochastic model over a deterministic becomes beneficial as uncertainty in peak timing increases.

On the benefit of flexibility. The sensitivity analysis in Section 4.2 showed that DA and RT resources benefit from flexibility in different ways. DA resources are used primarily for peak reduction and are largely insensitive to start-time granularity, since their multi-hour interruption windows are long relative to the timing precision being restricted. RT resources are used primarily to manage ramping and are therefore more sensitive to that same restriction. This effect is too large to be explained by the capacity of any single RT resource and better attributed to the loss of the ability to

coordinate timing across many resources in order to attack specific ramps that breach the ramp limit as well as reducing the overall ramping profile. This distinction broke down under wind uncertainty, where both resource types became markedly more sensitive to start-time granularity, which is likely due to a mismatch between the archetypes' hourly-shaped design and the true, non-hour-aligned timing of real wind generation data. Interruption duration flexibility had a larger effect on the objective than start-time granularity in most cases, this is most significant in the spike-and-peak-shift case, where the largest allowed RT interruption length reduced the objective by 37%. Flexibility clearly matters, but not in the same way for DA and RT resource. Which type of flexibility helps, and how much, depends on the resource's role and on the kind of uncertainty it has to respond to.

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Appendices

A Start-time Granularity

A.1 DA Results

Case	Metric	5-min	15-min	Half Hour	Hourly	Exactly on the Hour
Base Case	Objective	2731.39	2733.91	2730.08	2728.05	2734.34
	Ramping	1504.16	1515.74	1500.95	1512.59	1508.15
	Peak	1209.77	1211.23	1221.87	1215.46	1224.08
	Max 3-period Ramp	34.26	33.24	33.62	33.88	33.83
Net Load (Wind)	Objective	4459.93	4511.42	4739.26	4815.00	4630.18
	Ramping	3213.51	3210.73	3244.43	3249.09	3224.19
	Peak	675.12	687.11	708.23	723.05	708.15
	Max 3-period Ramp	71.02	72.02	72.48	72.81	69.83
AI Spike	Objective	2720.53	2736.36	2716.78	2723.62	2759.69
	Ramping	1467.27	1479.47	1475.51	1472.93	1481.36
	Peak	1210.74	1228.18	1220.32	1221.43	1226.12
	Max 3-period Ramp	32.48	32.97	34.50	34.38	33.44
Spike + Peak Shift	Objective	2741.01	2730.72	2757.15	2725.77	2749.96
	Ramping	1514.11	1489.03	1531.18	1498.45	1506.89
	Peak	1221.30	1225.16	1225.97	1227.31	1238.76
	Max 3-period Ramp	34.65	34.45	34.49	34.02	33.95

Table 3: Effect of restricting DA DR start-time granularity on objective, peak, and ramping across uncertainty cases (Base, Net Load with Wind, AI Spike, and Spike + Peak Shift). Columns range from unrestricted 5-minute placement to fully fixed, on-the-hour start times.

A.2 Real Time

Case	Metric	5-min	15-min	Half Hour	Hourly	Exactly on the Hour
Base Case	Objective	2681.87	2698.07	2727.08	2740.97	2722.54
	Ramping	1470.10	1484.55	1494.21	1512.47	1499.76
	Peak	1211.77	1209.65	1216.62	1215.85	1216.06
	Max 3-period Ramp	33.20	34.45	34.67	33.27	33.30
Net Load (Wind)	Objective	3933.21	4189.41	4255.65	4501.50	4505.75
	Ramping	3272.63	3351.08	3328.54	3330.43	3363.81
	Peak	638.92	649.76	654.12	659.55	672.87
	Max 3-period Ramp	69.68	72.92	74.23	77.39	73.69
AI Spike	Objective	2722.55	2717.68	2740.73	2764.11	2840.54
	Ramping	1464.04	1489.90	1499.49	1512.70	1515.80
	Peak	1218.26	1219.97	1212.28	1221.50	1218.84
	Max 3-period Ramp	34.71	34.82	34.48	34.32	35.90
Spike + Peak Shift	Objective	2743.81	2751.50	2774.74	2783.84	2771.97
	Ramping	1527.12	1529.22	1554.67	1539.04	1556.59
	Peak	1216.69	1217.84	1220.07	1225.17	1215.00
	Max 3-period Ramp	32.97	33.94	33.96	34.97	33.45

Table 4: Effect of restricting RT DR start-time granularity on objective, peak, and ramping across uncertainty cases (Base, Net Load with Wind, AI Spike, and Spike + Peak Shift). Columns range from unrestricted 5-minute placement to fully fixed, on-the-hour start times.

B Interruptible length

B.1 DA results

Case	Metric	1	1.5	2	3	4	5	6	7
Base Case	Objective	2666.81	2672.19	2666.13	2673.43	2660.53	2631.85	2627.40	2635.06
	Ramping	1432.36	1446.19	1433.91	1441.38	1436.93	1429.70	1432.33	1439.90
	Peak	1234.45	1225.00	1232.23	1232.05	1223.60	1202.14	1195.06	1194.41
Net Load (Wind)	Objective	4542.56	4556.10	4415.83	4512.88	4503.82	4493.76	4469.34	4419.40
	Ramping	3129.02	3117.64	3080.71	3117.68	3112.96	3123.09	3112.89	3111.89
	Peak	690.45	684.33	684.10	684.86	688.46	673.91	685.17	670.25
AI Spike	Objective	4646.76	4861.52	4367.71	4599.90	4553.11	4514.26	4534.35	4571.70
	Ramping	1680.76	1724.89	1678.32	1685.75	1704.86	1763.71	1715.62	1643.02
	Peak	1266.51	1270.12	1269.92	1276.64	1270.24	1270.59	1258.45	1256.31
Spike + Peak Shift	Objective	2907.23	2911.86	2909.05	3004.64	2895.03	2939.57	2911.08	2902.88
	Ramping	1597.64	1618.26	1607.49	1660.06	1597.21	1612.41	1587.91	1620.41
	Peak	1256.38	1254.81	1249.42	1258.40	1242.75	1241.55	1238.80	1243.27

Table 5: DA interrupt-length flexibility sensitivity across uncertainty cases (Base, Net Load with Wind, AI Spike, and Spike + Peak Shift).

B.2 RT Results

Case	Metric	1	1.5	2	2.5	3
Base Case	Objective	2725.16	2700.76	2695.32	2674.02	2640.67
	Ramping	1515.06	1482.51	1481.11	1460.00	1442.82
	Peak	1210.10	1215.96	1209.52	1214.00	1197.85
Net Load (Wind)	Objective	4058.28	4047.23	4010.32	4013.02	4003.87
	Ramping	3263.99	3263.87	3277.21	3267.00	3251.00
	Peak	664.81	663.28	664.52	664.00	661.49
AI Spike	Objective	2998.92	2944.11	2895.88	2875.67	2891.77
	Ramping	1634.84	1624.17	1601.71	1595.70	1615.18
	Peak	1252.11	1244.36	1232.11	1232.19	1232.64
Spike + Peak Shift	Objective	4471.92	4287.91	3524.19	3412.15	2813.66
	Ramping	1765.84	1762.59	1695.71	1701.32	1553.78
	Peak	1251.11	1251.64	1230.11	1223.55	1222.64

Table 6: RT interrupt-length flexibility sensitivity across uncertainty cases (Base, Net Load with Wind, AI Spike, and Spike + Peak Shift).

C Github Link

<https://github.com/michaeltiller/OR-Masters-Dissertation>